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AI for Bharat Hackathon

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Problem Statement : AI for Communities, Access & Public Impact.

Brief about the Idea:

The proposed solution is an AI-powered public assistance platform designed to improve access to public services for communities, especially in rural areas and among senior citizens. The platform integrates an AI Complaint Resolver to help citizens easily report local civic issues such as road damage, water supply problems, garbage, and electricity failures, and a Language Barrier Breaker to simplify government forms, notices, and scheme-related information in regional languages.

Using artificial intelligence, the system automatically understands user inputs, routes complaints to the appropriate authorities, provides transparent status tracking, and offers simplified text or voice-based explanations of government information. This ensures inclusive access, reduces dependency on middlemen, and improves transparency and efficiency in public service delivery.

Your solution should be able to explain the following:

- **How different is it from any of the other existing ideas?**

Unlike traditional complaint systems that require users to understand complex procedures and identify the correct government department, our solution uses AI to automatically analyze complaints and documents submitted through text, voice, or images. The system intelligently classifies the issue and routes it to the appropriate authority, making the process simple, fast, and accessible even for rural users and senior citizens.

- **How will it solve the problem:**

The platform simplifies civic complaint reporting and access to government information by providing a multilingual, voice-enabled interface. Users can submit complaints or upload documents in their regional language, while AI handles understanding, processing, and tracking. Real-time status updates and notifications ensure transparency and reduce delays in resolution.

- **USP (Unique Selling Proposition):**

A single AI-powered platform that integrates civic complaint resolution and language accessibility, enabling inclusive, transparent, and efficient public service delivery for communities, especially in rural areas and among senior citizens.

List of features offered by the solution:

- **AI-based Complaint Classification:**

Automatically understands civic issues from text, voice, or image inputs.

- **Automatic Department Routing:**

Routes complaints to the correct government authority without user effort.

- **Multilingual & Voice Support:**

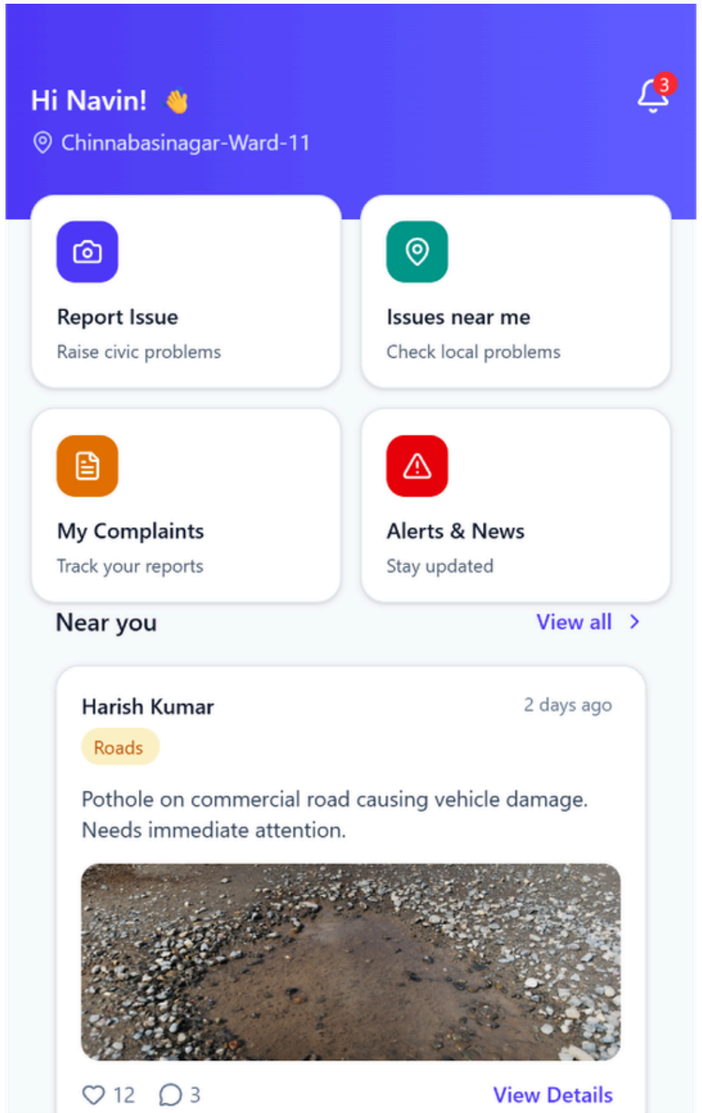
Allows users to interact in regional languages with voice assistance.

- **Government Document Simplification:**

Translates and explains government forms, notices, and schemes in simple language.

- **Real-time Status Tracking & Alerts:**

Provides transparent updates, reminders, and escalation notifications.

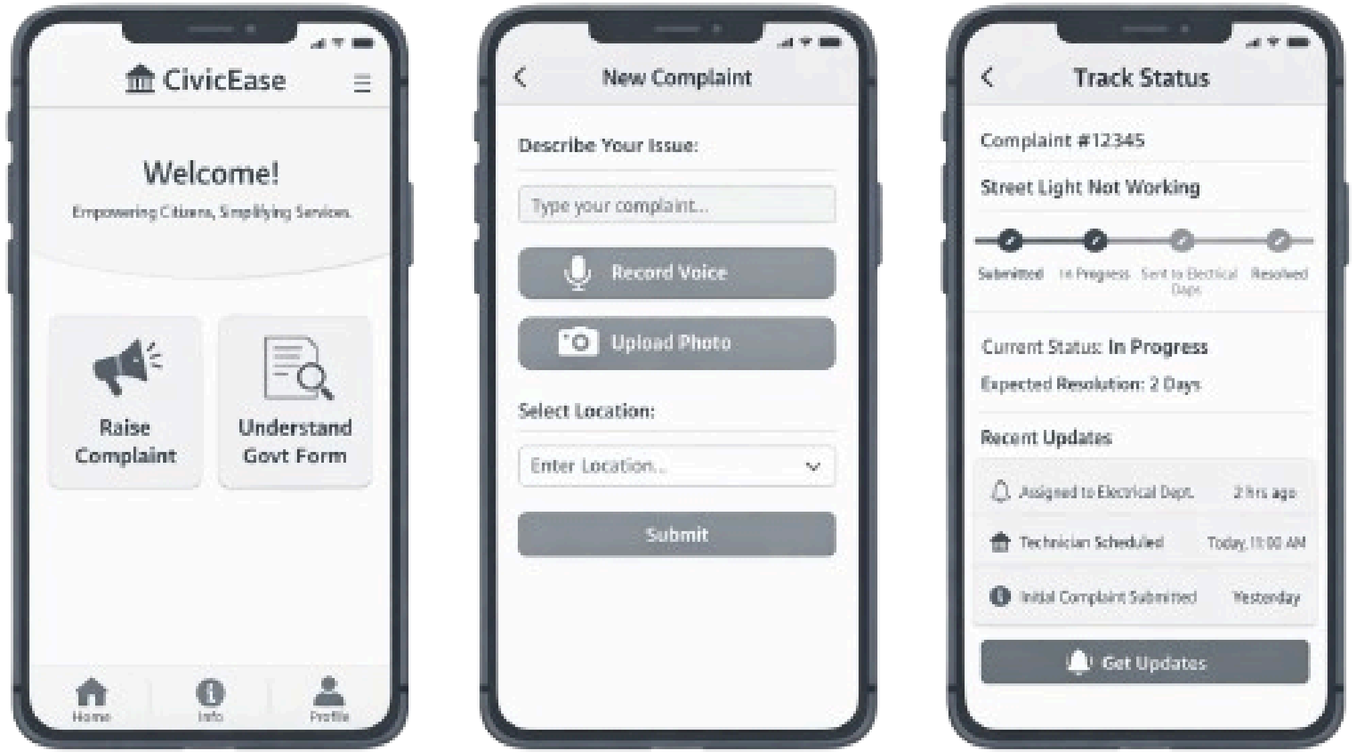


(“Images are used only for illustration;
the final prototype will be uniquely designed and implemented.”)

Process flow diagram or Use-case diagram:

User

→ Home Screen
→ Raise Complaint
→ Text / Voice / Image Input
→ AI Processing
→ Issue Classification
→ Department Routing
→ Status Tracking
→ Resolution & Feedback



(This flow shows how AI converts user inputs into structured complaints and ensures automatic routing and transparent tracking).

Wireframes/Mock diagrams of the proposed solution (optional):

Wireframes / Mock Diagrams of the Proposed Solution (Optional)
This slide showcases low-fidelity wireframes of the proposed AI-powered public assistance application. The mock diagrams represent the basic layout and user flow of the system and are created to visualize how users will interact with the platform.

Wireframes Included:

- Home Screen: Options to raise a civic complaint or understand government forms
- Complaint Submission Screen: Text, voice, or image-based input with location details
- Status Tracking Screen: Complaint ID with real-time status updates

These wireframes are conceptual and intended to demonstrate functionality and user experience. The final application interface will be developed as a unique AI-driven prototype.

Architecture diagram of the proposed solution:

Architecture Components:

- **Frontend (Mobile/Web App):**

User interface for complaint submission, document uploded and status tracking

- **AI Processing Layer:**

NLP for text, Speech-to-Text for voice Image Processing for photos Language Translation

- **Backend Services:**

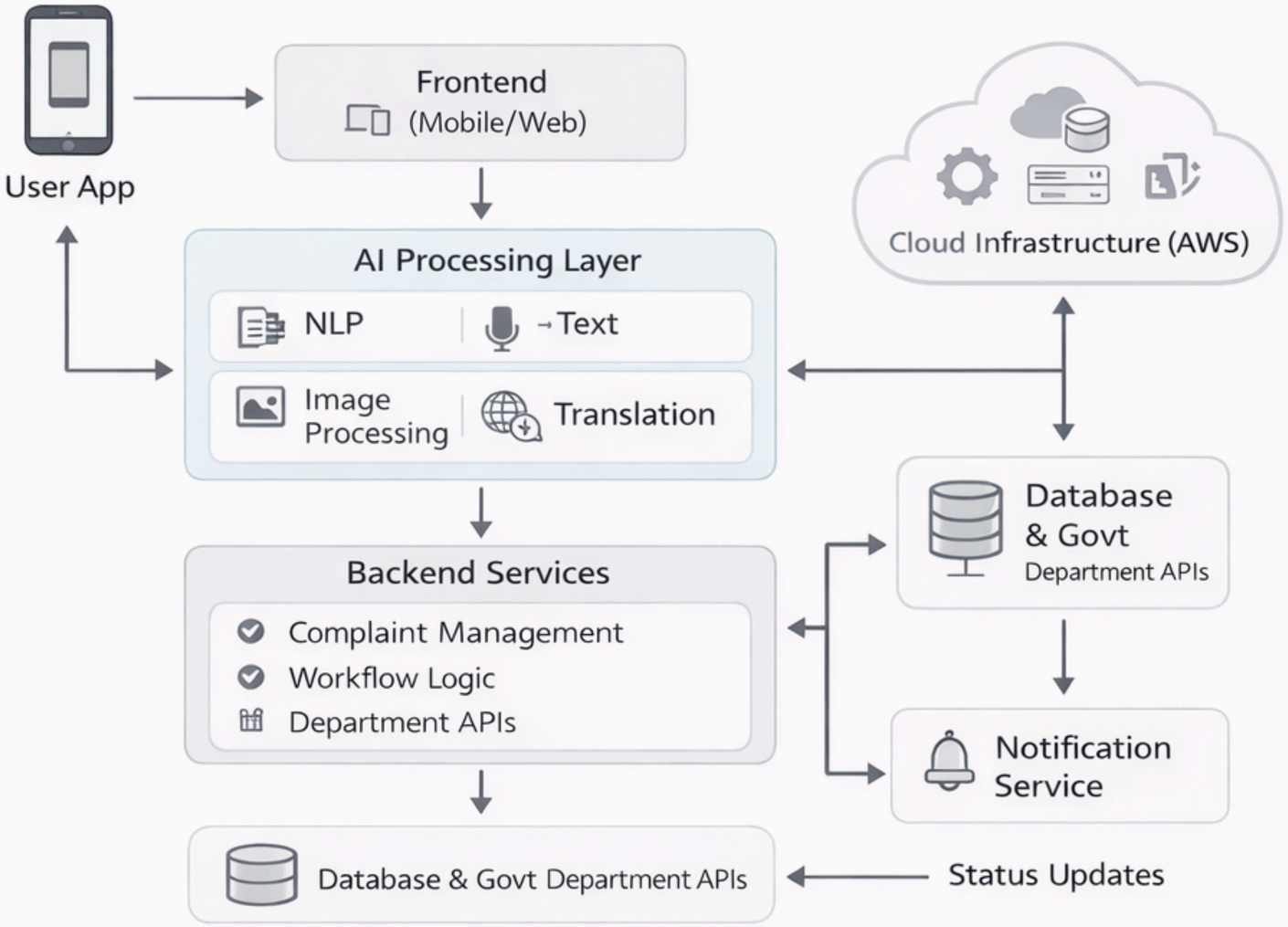
Complaint management, workflow logic, department routing, notifications

- **Database**

Stores user data, complaints, documernt, status updates

- **Cloud Infrastructure (AWS)**

Scalable computing, storage, API services



This architecture demonstrates how AI-driven processing enables automated complaint routing, language support, and real-time public service delivery.

Technologies to be used in the solution:

Multilingual Translation: For enabling regional language support and inclusive access.
Text-to-Speech (TTS): For providing voice responses and status updates to users.
Backend Framework (Flask/FastAPI): For handling APIs, business logic, and AI integration.
Frontend Technology (Web/Mobile App): For providing a simple, accessible user interface.
Database Management System: For storing user data, complaints, and tracking status securely.
Cloud Platform: For scalable deployment, storage, and real-time availability.
Optical Character Recognition (OCR): For extracting text from uploaded documents and images submitted by users.
Intent Recognition Engine: For identifying the purpose of user requests and complaints automatically.
Auto Department Mapping Logic: For linking classified complaints to the appropriate government authority.
Geolocation Services: For capturing the exact location of reported issues to enable faster resolution.
Notification Services: For sending real-time updates via SMS, app alerts, or email.

Estimated implementation cost (optional):

Cloud Infrastructure (AWS):
Uses student/free tier and pay-as-you-go services for compute, storage, and APIs.

AI & Language Services:
Open-source models and limited API usage to reduce operational cost.

Development & Maintenance:
Student-led development with minimal maintenance cost.

Overall Cost:
Low-cost, scalable, and suitable for pilot-level implementation.

Add as per the requirements for the hackathon:

Impact:

- Enables faster and more transparent resolution of civic complaints
- Improves accessibility to public services for rural communities
- Reduces dependency on manual processes and intermediaries
- Builds trust between citizens and government authorities
- Provides inclusive support for senior citizens and non-technical users

Future Scope:

- Integration with official government portals and municipal systems
- Expansion to support additional regional languages
- AI-based priority assignment and analytics dashboard for authorities
- Offline access through SMS or local service kiosks
- Potential adoption across multiple cities and rural regions

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Thank You